THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — MATHS · HARD LEVEL CHAPTER TEST SET 1

Chapter 8: **Sequences and Progressions** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The 6th term of an AP is 20 and the 10th term is 32. The 16th term is: 1

(a) 50 **(b)** 47 **(c)** 56 **(d)** 53

2. The sum of the first n terms of a sequence is $S_n = 2n^2 + 3n$. Its 9th term is: 1

(a) 37 **(b)** 39 **(c)** 41 **(d)** 35

3. The sum of all integers from 1 to 100 that are divisible by 6 is: 1

(a) 810 **(b)** 826 **(c)** 816 **(d)** 822

4. The sum of the first 12 terms of an AP is 510 and its first term is 4. The common difference is: 1

(a) 7 **(b)** 6 **(c)** 8 **(d)** 9

5. Three numbers in AP have sum 30 and product 840 (with $d > 0$). The largest number is: 1

(a) 15 **(b)** 14 **(c)** 10 **(d)** 18

6. A town's population is 10000 and grows by 10% each year. After 2 years it is: 1

(a) 12100 **(b)** 12200 **(c)** 12000 **(d)** 12300

7. The first term of a GP is 3 and the 8th term is 384 (with $r > 0$). The common ratio is: 1

(a) 1 **(b)** 2 **(c)** 4 **(d)** 3

8. The nth term of an AP is $t_n = 3n + 2$. The sum of its first 16 terms is: 1

(a) 443 **(b)** 437 **(c)** 456 **(d)** 440

9. **Assertion (A):** The sequence $2, 5, 8, 11, \ldots$ is an arithmetic progression. 1
Reason (R): The difference between consecutive terms is the constant 3.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** The sequence $2, 4, 8, 16, \ldots$ is a geometric progression. 1
Reason (R): The sequence $1, 3, 5, 7, \ldots$ is an arithmetic progression.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. The sum of the first 15 terms of an AP is 255 and the first term is 3. Find d and the 15th term. 2

12. Which term of the AP $5, 11, 17, \ldots$ is the first term to exceed 200? 2

13. If $S_n = 3n^2 + 4n$ is the sum of the first n terms of a sequence, find its first term and its 10th term. 2

14. The sum of the first 12 terms of an AP is 336 and the first term is 50. Find d and the 12th term. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The sum of the 3rd and 9th terms of an AP is 40 and its 5th term is 17. Find the AP's first term and the sum of its first 20 terms. 3

16. Find the sum of all natural numbers between 10 and 200 that leave remainder 2 when divided by 5. 3

17. Three positive numbers are in GP. Their product is 216 and the sum of the first and third is 15. Find the numbers. 3

OR

The sum of the 2nd and 10th terms of an AP is 64 and its 5th term is 28. Find the AP's first term and the sum of its first 14 terms. 3

18. Find the sum of all natural numbers between 20 and 300 that leave remainder 3 when divided by 7. 3

OR

Three positive numbers are in GP. Their product is 1728 and the sum of the first and third is 40. Find the numbers. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Meera starts a recurring savings plan in a bank. She deposits Rs 100 in month 1, Rs 150 in month 2, Rs 200 in month 3, and so on, increasing the deposit by the same amount every month. 5

(i) Write the first term and the common difference.

(ii) Find the deposit in month 12.

(iii) Find the total amount deposited in 12 months.

(iv) In which month is the deposit Rs 1000?

20. A contractor pays a penalty for delay: ₹100 on day 1, ₹125 on day 2, ₹150 on day 3 and so on. If the work is 24 days late, find the penalty on the last day and the total penalty. 5

OR

A ball is dropped from 64 m and rebounds to 3/4 of the previous height. Show that the rebound heights form a GP and find the 3rd rebound height. 5

END OF QUESTION PAPER — Total: 40 Marks